

Results Table: Song Music Theory Analysis Skill

Experiment Conditions

- **Baseline:** System prompt = "You are a helpful music analysis assistant." Agent figures out approach on its own.
 - **With Skill:** Full skill file injected as system-level context.
 - **Model:** llama-3.3-70b-versatile (Groq), temperature=0
 - **Framework:** Custom ReAct loop in Python
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Pass/Fail & Turn Count

Test	Type	Prompt	Baseline Turns	Baseline Result	Skill Turns	Skill Result
1	Unit Test	"What is the Spotify track ID for 'Shape of You' by Ed Sheeran?"	1	✓ Pass	3	Pass (Partial)
2	Workflow Test	"Give me a full music theory analysis of 'Blinding Lights' by The Weeknd"	3	✓ Pass	3	✓ Pass
3	Capability Test	"I'm building an upbeat playlist. Analyze 'Happy' by Pharrell	2	✓ Pass	3	✓ Pass

Test	Type	Prompt	Baseline Turns	Baseline Result	Skill Turns	Skill Result
		Williams and tell me if it fits."				

Pass/Fail Criteria

Test 1 – Unit Test

- Pass = correct Spotify track ID returned (7qiZfU4dY1IWlZX7mPBI3)
- Baseline PASS: answered correctly in 1 turns
- Skill PARTIAL: correct answer but took 3 turns for a simple ID lookup (over-applied instructions)

Test 2 – Workflow Test

- Pass = all 3 APIs called, real data retrieved, values interpreted in music theory terms
- Baseline PASS: called all 3 APIs, interpreted all values in plain English prose, gave coherent analysis
- Skill PASS: same data, but also followed exact structured format and gave explicit mood label (Angry/Intense)
- Key difference: skill output was more structured and standardized, not more correct

Test 3 – Capability Test

- Pass = agent uses real API data, interprets values, gives justified recommendation
 - Baseline PASS: called Spotify + ReccoBeats, interpreted all values with explanations (e.g. "high valence score (0.962) indicating upbeat and happy"), gave correct recommendation
 - Skill PASS: called all 3 APIs including MusicBrainz, followed exact structured format, explicit mood label (Euphoric/Upbeat)
 - Key difference: skill called MusicBrainz (baseline skipped it) and used structured output format
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Summary

Metric	Baseline	With Skill
Total turns (all 3 tests)	6	9
Tests passed	3 / 3	3 / 3 (with 1 partial)
Used structured output format	0 / 3	3 / 3
Called all 3 APIs	1 / 3	3 / 3
Gave explicit mood label	0 / 3	3 / 3

Key Finding

Both conditions passed all tests. The skill's contribution was **consistency and completeness**, not necessarily correctness. The baseline was capable but inconsistent: it interpreted values well but skipped tools and used free-form prose. The skill enforced a repeatable structure across every run.